

SUBHRARTHA CHATTERJEE

GIS ANALYST

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SUMMARY

Detail-oriented GIS Analyst with hands-on experience in GIS support for electrical distribution networks within DISCOM projects. Experience in field surveys, GPS data capture, feeder QC, PSS/GSS digitization, and incorporation of built-in drawings into GIS systems. Skilled in maintaining network asset data, validating spatial and attribute information, and supporting SCADA-integrated GIS applications for operational planning and monitoring. Builds custom ArcGIS Python (ArcPy) automation tools, with AI-assisted development, to streamline large-scale geodatabase data entry, validation, and QC workflows.

SKILLS

GIS Operations: Spatial Data Creation, Editing, and Analysis

GIS Software: ArcGIS, QGIS, AutoCAD, GE Smallworld, Google Earth Pro

Data Management: Attribute Tables, Geodatabases, Data Validation

Automation & Scripting: Python (ArcPy) Toolbox Development, AI-Assisted Tool Building, Workflow Automation

Mapping & Visualization: Thematic Mapping, Layout Design, Spatial Reporting using Advance Excel

PROFESSIONAL EXPERIENCE

GIS ANALYST | *NIYAL TRITECH PVT LTD*

April 2026 – Present

- **Performing field data collection, feeder QC, and PSS/GSS digitization** using ArcGIS and AutoCAD to build and maintain accurate electrical distribution network datasets.
- **Developed custom ArcGIS Python (ArcPy) toolboxes with AI-assisted development** to automate recurring GIS data workflows, cutting manual data-entry effort across large multi-layer geodatabases.
- Built a bulk attribute-fill tool (40+ shared fields across all layers in one run) and a geodatabase QC tool validating mandatory fields and domain-code compliance across 50+ feature classes, producing a single consolidated CSV report.
- Built a survey ID generation and spatial linking tool (auto-generated IDs, feature codes, UTM coordinates, line-to-structure linking) and an Excel-to-geodatabase loader with automatic domain-code correction.

GIS EXECUTIVE | *TATA POWER SOUTHERN ODISHA DISTRIBUTION LTD*

May 2024 – March 2026

- **Surveying electrical network assets** such as poles, RMU, distribution transformers, cables, conductors, and consumers using DGPS to capture coordinates and create a structured GIS database.
- **Digitization of base maps and electrical distribution networks** using satellite imagery and Georeferencing techniques.
- **Supporting route planning** for new electrical lines, network expansion, and maintenance activities using geospatial analysis.
- **Conducting field verification** and coordinating with survey teams for ground-truthing of electrical network assets.
- **Supported electrical distribution modernization projects** using GE Smallworld, ArcGIS, QGIS, and AutoCAD for GIS-based network mapping.
- Collecting and updating consumer details through household surveys, including codification and indexing with their respective power supply sources.
- Ensuring synchronization of GIS asset data (feeders, distribution transformers, switches, substations, and HT/LT lines) with ADMS for accurate electrical network modeling.
- Coordinating with Operations, Metering, STS, Projects, and DOS teams to support data integration and system alignment.
- Maintaining data accuracy, validation, and quality control of GIS databases to support energy auditing, AT&C loss reduction, and network planning.

- Preparing land base data for GIS mapping and electrical network modeling.
- Conducting field surveys using mobile applications to capture substations and electrical assets, including HT/LT poles, HT/LT lines or cables, and distribution transformers.
- Creating and digitizing electrical distribution networks using ArcMap 10.8, QGIS, and Google Earth.
- Performing consumer tagging and linking consumers with their respective electrical network assets.
- Performing data quality checks (QC), validation, and corrections to finalize GIS datasets.

INTERNSHIP

TECHNICIAN TRAINEE

October 2022 – December 2023*Indorama India Pvt. Ltd.*

- Performed maintenance and troubleshooting of motors, motor control centers (MCC), transformers, circuit breakers, and diesel generators (DG).
- Installed and maintained star-delta, DOL, and forward/reverse motor starters, carrying out motor connection/disconnection per safety procedures.
- Inspected transformers, circuit breakers, and protective relays; operated synchronizing panels for grid-to-STG (steam turbine generator) synchronization.
- Read, interpreted, and prepared single line diagrams (SLD) for electrical systems.

EDUCATION

Bachelor of Technology in Electrical Engineering

Expected: 2027*Maulana Abul Kalam Azad University of Technology, West Bengal*

Diploma in Electrical Engineering

Pass Out: 2020*West Bengal State Council of Technical & Vocational Education and Skill Development, West Bengal**CGPA: 7.05 / 10*

Higher Secondary (HS) Examination in Science

Pass Out: 2017*West Bengal Council of Higher Secondary Education, West Bengal**CGPA: 6.65 / 10*

Secondary Examination in General Study

Pass Out: 2010*West Bengal Board of Secondary Education**CGPA: 5.14 / 10*

CERTIFICATIONS

- **Electrical Supervisor Certificate – West Bengal Electrical Licensing Board (2024):** S.S.C No. SCC02/2024/01111; parts qualified: 1, 2, 11; issued on 05/08/2024; valid up to 05/08/2034.
- **Information Technology Application Certificate – Department of Youth Services, Government of West Bengal (2022):** Certificate No. WB1136115221; grade: A+; duration: Jan 2022 – Jun 2022; issued on 09/11/2022.

PERSONAL DETAILS

- **Father's name:** Subir Chatterjee **Mother's name:** Shikha Chatterjee
- **Address:** Vill + P.O – Majurdanga, P.S – Patrasayer, District: Bankura, West Bengal, PIN: 722206
- **Date of birth:** 16/10/1998 **Nationality:** Indian **Caste:** General

STRENGTHS

- Strong multitasking abilities
- Good problem-solving skills
- Excellent communication skills
- Active listening and teamwork
- Self-motivated and responsible

Declaration

The information provided herein is true and correct to the best of my knowledge and belief.